

Leo Bogaert

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EDUCATION

Western University

Bachelors of Engineering Science, Electrical Engineering (B.E.Sc.)

- GPA: **3.9/4.0**; Dean's Honour List

April 2029

London, ON

HARDWARE PROJECTS

Custom Class D Amplifier

- Designed a custom Class-D audio amplifier PCB to drive a 4 Ω speaker using a MOSFET switching stage
- Utilized a 555 timer and comparator converting analog audio into a digital PWM signal for the amplifier's MOSFET stage.
- Designed the audio signal path, output filtering, and power-supply decoupling for clear and reliable audio reproduction while selecting mass produced components based on voltage, current, switching, and power requirements.
- Developed and prepared a 2-layer PCB in Altium Designer for fabrication, verifying that the design followed all JLCPCB manufacturing rules using DRC and custom set rules

STM32 Based Flight Controller Board

- Designed a flight controller PCB based around an STM32F446, with IMU, barometer, USB, SWD programming, and nine PWM output modules.
- Configured a 4-layer stack-up and routed USB differential pairs to a controlled impedance of approximately 90 Ω while applying grounding, return-path, and power-distribution considerations.
- Utilized SPI/I2C, 8 MHz crystal oscillator, and integrated 3.3V/5V regulation to power the board

3D Printed RC Drone

- Developed a high-strength, rigid drone frame using Fusion 360, printed out of carbon-filled PETG.
- Performed iterative flight testing to validate structural rigidity and tune control responsiveness using Betaflight to achieve stable, responsive flight performance
- Interfaced an RC receiver with the flight controller over UART, enabling wireless command and control.

EXPERIENCE

Western WEBOTS

Low Voltage Electrical & Mechanical Member

- Designed a custom harmonic drive using onshape with a 20:1 torque ratio, implemented as a hip joint for humanoid robots.
- Designed a central controller utilizing a NVIDIA Jetson Orin Nano to run the autonomous navigation systems
- Configured a high speed brushless motor driver to enable movement of robot joints

Sep. 2025 – April 2026

London, ON

Mandarin

Host

- Built advanced communication skills, consistently conveying information clearly between customers, and management
- Assisted management by training new employees in phone operations, dishes and hosting procedures

Apr. 2023 – Sep 2025

London, ON

Enders Direct

Computer Hardware Technician

- Studied under a senior technician, developing practical skills in troubleshooting and repair methodology.
- Built a strong competency in identifying and resolving computer hardware faults and software-related issues

Sep. 2023 - Feb 2024

London, ON

Technical Skills

- **Electronics & PCB:** Altium Designer, KiCad, Analog/Digital/Power Circuits
- **Embedded & Communication:** ESP32, STM32, UART, SPI, I2C, PWM
- **Programming:** C++, C# Python, Java
- **CAD & Development:** Fusion 360, OnShape, Git, GitHub, Bambu/Prusa Slicer
- **Lab Equipment:** Oscilloscope, Digital Multimeter, Function Generator, Soldering, Serial Debugging